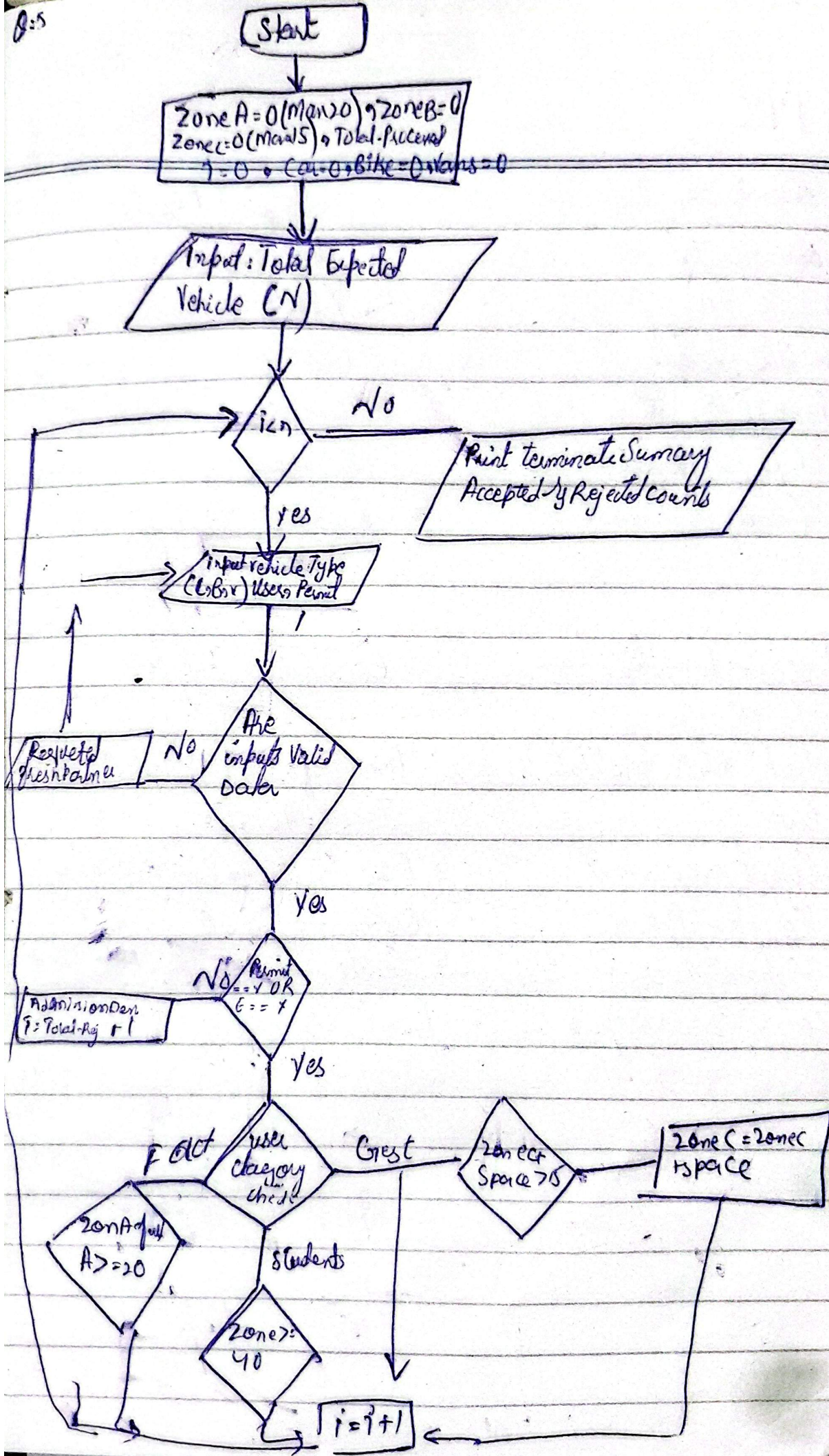


Q:5



① PAC

Given Data

- Total expected Vehicle (N)
- Vehicle details
Type (C/B/V), User (F/S/G), Permit (Y/N)

Required Results

- Parking Confirmation (Zone A/B or C)
- Error
- Terminal Summary Report (Occupancy %s, counts, break down)

Processing Formulas:

- Check $i < N$ & validate Data
- Count increment
Cars++, Bike++
- Capacity Tracking
 $Zone = Zone + space$

Constraints / Condition

- Max Capacities Zone A = 20,
Zone B = 40, Zone C = 15
- Access priority → Zone A (overflows)
Student → Zone B
Guest → Zone C
Permit or Emergency

IPO

Input	Processing	Output
• N (Total Vehicles) • Type (C/B/V) • Use (F/S/G) • Permit (Y/N)	1. Initialize all counter & zones to 0 2. Loop N times to input & vehicle data 3. Check permit/emergency status 4. Route user to assigned zone & check capacity 5. Increment repetitive counter & final summary	• Parking Confirmation status • Denial notification • Final Terminal Summary

Algorithm:

- 1 Start by Initialize ($\text{ZoneA}=0, \text{ZoneB}=0, \text{ZoneC}=0$), counts
Total-Processed = 0, Total-Rejected = 0, Cars = 0, Bikes = 0
by loop counter $i=0$
- 2 Input total expected Vehicles N
- 3 loop if $i < N$, Processed to step 4; otherwise go to step 8
- 4 Input Vehicle details Type, User, Permit
- 5 Validate data: If invalid, Print error by Repeat step 4
- 6 If Permit $1 = Y$ by Emergency = Y increment = Total-Rejected
increment = i by return to step 3
- 7 Allocate Zone based on User.
 - $F \Rightarrow$ If $\text{ZoneA} < 20$, Park in Zone A. Else overflow to zone B
 - $S \rightarrow$ Go to Zone B check
 - $G \rightarrow$ If $\text{ZoneC} < 15$, Park in Zone C. Else, go to zone B
 - Zone B check: If $\text{ZoneB} < 40$, Park in Zone B, else Go to step 8
- 8 Hand Full Capacity = Total-Rejected - User Entered
Increment = $i = i + 1$
- 10 End

Pseudocode:

Start

Begin

Initialize Zone A = 0, Zone B = 0, Zone C = 0, Total-Rejected = 0,
Cars = 0, Bikes = 0

Input N

While $i < N$ Do

Repeat

Input Type, User, Permit, Emergency

If Input are Valid then

Print "Invalid Category Code Combination"

End If

Until Inputs are Valid

If Permit == Y OR Emergency == Y then

If User == 'F' then

If Zone A < 20 then

Zone A = Zone A + Space

Increment Category Counter

Print "Parked in Zone B"

Else

Total-Rejected = Total-Rejected + 1

Print "Rejection: Assigned Area Exhausted"

End If

Else If User == G then

If Zone C < 15 then

Zone C = Zone C + Space

Increment Category Counter

Print "Parked in Zone C"

Else

Total-Rejected = Total Rejected + 1

Print "Admission Denied: Missing Permit"

End If

$i = i + 1$

End While

Print Terminal-Summary-Report

End